ÖSTERREICHISCHE NATIONALE WAHLSTUDIE
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AUTNES Automatic Content Analysis of the Media Coverage 2017

Documentation

Version 1.1

Olga Litvyak, Andreas Fischeneder, Gregor Rettenegger, Sebastian Galyga, Jakob-Moritz Eberl, Wolfgang C. Müller, Sylvia Kritzinger, Hajo G. Boomgaarden

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Introduction

This Documentation Report accompanies and describes the data of the AUTNES Automatic Content Analysis of Media Coverage of the Austrian National Election held on 15 October 2017. The body elected was the National Parliament (Nationalrat).

The Documentation Report is structured in six sections. The first section outlines the conditions of use of the data. The second section, the study description, provides an overview of the study. The third section presents the study design, i.e., the media sample and the sampling procedure. The fourth section provides information about the coding procedure, presenting all the variables and the units of analysis. The sixth section describes the validation procedure. The last section provides an overview of the frequencies of all variables in the analyzed media outlets. Precise information about the coding instructions for the validation procedure can be found in the Validation Codebook (in German), which is also available via www.aussda.at.

Acknowledgement of the data

Data users are required to acknowledge the data and the accompanying release document. Please refer to the AUSSDA data catalogue (https://data.aussda.at, doi: 10.11587/UCQBQQ) for a recommendation on how to cite these data and the documentation.

1. Conditions of Use

1.1. Restrictions

The data are available under the terms listed at the AUSSDA Dataverse.

1.2. Confidentiality

AUTNES, the Principal Investigators and the funding institution bear no responsibility for the use of the data, or for interpretations or inferences based on their use, neither do they accept liability for indirect, consequential or incidental damages or losses arising from the use of the data.

1.3. Deposit Requirement

To facilitate exchanges within the scientific community and to provide funding agencies with essential information about the use of archival resources, users of the AUTNES data are requested to notify the AUTNES team of all forms of publications referring to AUTNES data.

2. Study description

2.1. Title and Edition

AUTNES Automatic Content Analysis of the Media Coverage 2017

2.2. Principal Investigators

Wolfgang C. Müller, University of Vienna (Supply Side)

Sylvia Kritzinger, University of Vienna (Demand Side)

Hajo G. Boomgaarden, University of Vienna (Media Side)

2.3. Funding and Acknowledgements

The study was carried out under the auspices of the ACIER - Austrian Cooperative Infrastructure for Electoral Research and VieCER - Vienna Center for Electoral Research, financed by the Austrian Federal Ministry of Education, Science and Research. This data collection has been supported by the Austrian Press Agency (APA) via the APA-UNIVIE Data Project.

2.4. Data File Names

10753_da_en_v1_0.dta	Core data file - STATA format - 55 Variables, 46203 Observations
10753_da_en_v1_0.csv	Core data file - CSV format - 55 Variables, 46203 Observations
10753_da_en_v1_0.sav	Core data file - SPSS format - 55 Variables, 46203 Observations

2.5. Topics /Keywords

Election coverage, Political actors, Mass media coverage, Elections, Election campaigns, Political parties, Political representation, Political issues, European Union, Content Analysis

3. Study design

The data set is based on the automatic analysis of all political news reports published in Austrian media outlets (print, online, TV) leading up to the Austrian National Election 2017. We define a news report as political when the report mentions a central political actor. Central political actors are

- designations such as government (coalition, governing parties) and opposition (opposition parties) on the national level
- members of the government before the election, on national, state and EU level
- ministries and state secretaries (Staatssekretäre) on the national level
- the Austrian President
- members of parliament before the election, both national and EU level, or the parliament as an institution
- Austrian political parties eligible on the national level, including demographically defined sub-organizations on the national level (e.g. for women)
- directorate members of demographically defined sub-organizations of Austrian parties (also political fractions in unions)
- political candidates running for parliament or office in the election

If any of these actors is mentioned in a news item, it was considered political news. The period of analysis ranges from 6 June to 15 October 2017 for print and TV media, and from 4 August to 15 October 2017 for online media. Note that the time range of news reports extends the range covered by the AUTNES manual content analysis. The media data can therefore be synced with AUTNES demand side data, specifically with survey waves 1-4 of the AUTNES Online Panel Study as well as waves 1 and 2 of the AUTNES Multi-Mode Panel Study (see AUTNES Online Panel Study 2017, doi:10.11587/I7QIYJ and AUTNES Multi-Mode

Panel Study 2017, doi:10.11587/NXDDPE for more information on and documentation for the survey data).

3.1. Media Sample

The media sample was selected based on the type and circulation to reflect the Austrian national media landscape. All media reports that mentioned a relevant political actor were downloaded from the APA (Austrian Press Agency) database. Political actors in that sense are all members of the government, all lead candidates, and the parties themselves that were represented in the parliament before or after the elections. The data includes reports from 21 media outlets, covering print media, TV, and online media. For an overview see *Table 1*.

Table 1. Media sample and characteristics of media outlets.

Media outlet	Time	Media	Type	N
ZiB1	06.06.-15.10.	Public Service TV	TV	370
ZiB2	06.06.-15.10.	Public Service TV	TV	305
ATV	06.06.-15.10.	Commercial TV	TV	245
Pro7 – Puls 4	06.06.-15.10.	Commercial TV	TV	343
Der Standard	06.06.-15.10.	Quality Print	Print	2,770
Die Presse & Die Presse am Sonntag	06.06.-15.10.	Quality Print	Print	2,692
Salzburger Nachrichten	06.06.-15.10.	Quality Print	Print	2,152
Kronen Zeitung	06.06.-15.10.	Tabloid Print	Print	5,757
Österreich	06.06.-15.10.	Tabloid Print	Print	4,533
Heute	06.06.-15.10.	Tabloid Print	Print	1,386
Kurier	06.06.-15.10.	Mid-market Print	Print	4,881
Kleine Zeitung	06.06.-15.10.	Mid-market Print	Print	3,839
derstandard.at	04.08.-15.10.	Quality online	Online	2,061
diepresse.com	04.08.-15.10.	Quality online	Online	1,807
salzburg.com/sn.at	04.08.-15.10.	Quality online	Online	1,186
krone.at	04.08.-15.10.	Tabloid online	Online	989
oe24.at	04.08.-15.10.	Tabloid online	Online	1,328
heute.at	04.08.-15.10.	Tabloid online	Online	1,042
orf.at	04.08.-15.10.	Mid-market online	Online	4,509
kurier.at	04.08.-15.10.	Mid-market online	Online	1,541
kleinezeitung.at	04.08.-15.10.	Mid-market online	Online	2,467

3.2. Sample of News Items

To collect the data, articles for print newspapers were downloaded from the database of the Austria Presse Agentur (APA, www.apa-defacto.at), using the following search string containing the names of all Austrian political parties participating in the National Election of 2017, the names of the chancellor, all ministers and secretaries of state, as well as the national parliament as an institution. The search string that was used to retrieve those articles was:

```
nationalrat* OR spö* OR övp* OR "liste kurz"~4 OR "neue* volkspartei" OR fpö* OR neos* OR "liste pilz"~4 OR "kurz övp"~5 OR "sebastian kurz"~4 OR "minister kurz"~5 OR "bundesminister* kurz"~5 OR "europa* kurz"~5 OR "integration* kurz"~5 OR "außenminister* kurz"~5 OR "äußeres* kurz"~5 OR "kurz volkspartei"~5 OR "kurz vp"~5 OR "kanzler* kern"~5 "bundeskanzler* kern"~5 "christian kern"~4 "kern spö"~5 "kern sozialdemokrat*"~5 OR "minister* stöger"~5 OR "bundesminister* stöger"~5 OR "gesundheit* stöger"~5 OR "sozial* stöger"~5 "arbeit* stöger"~5 OR "alois stöger"~4 OR "brandstetter minister*"~5 OR "bundesminister* brandstetter"~5 OR "wolfgang brandstetter"~4 OR "justiz* brandstetter"~5 OR doskozil OR drozda OR "minister* hammerschmid"~5 OR "bundesminister* hammerschmid"~5 OR "bildung* hammerschmid"~5 OR "sonja hammerschmid"~4 OR karmasin OR "leichtfried minister*"~5 OR "verkehrsminister* leichtfried"~5 "jörg leichtfried"~4 OR "minister* mahrer"~5 OR "bundesminister* mahrer"~5 OR "wirtschaft* mahrer"~5 OR "wissenschaft* mahrer"~5 OR "forschung* mahrer"~6 OR "staatssekretär* mahrer"~5 OR "harald mahrer"~4 OR "minister* rendi wagner"~5 OR "minister* wagner"~5 OR "bundesminister* rendi wagner"~5 OR "bundesminister* wagner"~5 OR "gesundheitsminister* wagner"~5 OR "gesundheitsminister* rendi wagner"~5 OR "frauenminister* wagner"~5 OR "frauenminister* rendi wagner"~5 OR "pamela rendi wagner"~5 OR "rendi wagner"~5 OR "minister* rupprechter"~5 OR "andrä rupprechter"~4 OR "landwirtschaftsminister* rupprechter"~5 OR "umweltminister* rupprechter"~5 OR "minister* schelling"~5 OR "finanzminister* schelling"~5 OR "hans jörg schelling"~4 "hans jörg schelling" OR "johann georg schelling" OR ("minister* sobotka"~5 NOT tschechi*) OR "innenminister* sobotka"~5 OR "wolfgang sobotka"~4 OR "van der bellen" OR bundespräsident OR strache* OR "matthias strolz"~4 OR "neos* strolz"~6 OR lunacek OR "peter pilz"~4 OR "ex grün* pilz"~6 OR "grüne* pilz"~6 OR "politiker* pilz"~5 OR "aufdecker* pilz"~5 OR "untersuchung* pilz"~5 OR
```

```
"abgeordnete* pilz"~5 "parlament* pilz"~5 OR lunacek OR "rot grün" OR "schwarz grün" OR (("die grüne*" OR "der grüne*" OR "den grünen") NOT ("grün* rasen"~2 OR "grün* wiese*"~2)) OR ökopartei OR grünpolitiker* OR "grün* politiker*"~4 OR "tiroler grüne*" OR "grüne* tirol" OR "wiener grüne*" OR "grüne* wien" OR "grüne* burgenland" OR "burgenländ* grüne*" OR "grüne* kärnten" OR "kärntner* grüne*" OR "grüne* salzburg" OR "salzburger* grüne*" OR "grüne* oberösterreich" OR "oberösterreichische* grüne*" OR "grüne* steiermark" OR "steirische* grüne*" OR "grüne* vorarlberg" OR "vorarlberger grüne*" OR "grüne* niederösterreich*" OR "niederösterreichische* grüne*"
```

Articles were deduplicated based on text, title, date, medium (all of those had to be the same to be considered a duplicate). Due to the German national Election that took place around the same time in 2017, several articles reporting on it were identified by the initial search string. Face validity checks ensured the validity of this search string, as the number of articles reporting on the 2017 German national Election was not sufficient to apply any other validation procedure. The articles were first identified using the following search string:

```
(merkel* OR seehofer OR weidel OR gauland OR petry OR "simone peter" OR söder OR lindner OR özdemir OR schulz OR nahles OR spd OR cdu OR csu OR fdp OR afd OR bundestagswahl) NOT (nationalrat* OR spö* OR övp* OR "liste kurz"~4 OR "neue* volkspartei" OR fpö* OR neos* OR "liste pilz"~4 OR "kurz övp"~5 OR "sebastian kurz"~4 OR "minister kurz"~5 OR "bundesminister* kurz"~5 OR "europa* kurz"~5 OR "integration* kurz"~5 OR "außenminister* kurz"~5 OR "äußeres* kurz"~5 OR "kurz volkspartei"~5 OR "kurz vp"~5 OR "kanzler* kern"~5 "bundeskanzler* kern"~5 "christian kern"~4 "kern spö"~5 "kern sozialdemokrat*"~5 OR "minister* stöger"~5 OR "bundesminister* stöger"~5 OR "gesundheit* stöger"~5 OR "sozial* stöger"~5 "arbeit* stöger"~5 OR "alois stöger"~4 OR "brandstetter minister*"~5 OR "bundesminister* brandstetter"~5 OR "wolfgang brandstetter"~4 OR "justiz* brandstetter"~5 OR doskozil OR drozda OR "minister* hammerschmid"~5 OR "bundesminister* hammerschmid"~5 OR "bildung* hammerschmid"~5 OR "sonja hammerschmid"~4 OR karmasin OR "leichtfried minister*"~5 OR "verkehrsminister* leichtfried"~5 "jörg leichtfried"~4 OR "minister* mahrer"~5 OR "bundesminister* mahrer"~5 OR "wirtschaft* mahrer"~5 OR
```

```
"wissenschaft* mahrer"~5 OR "forschung* mahrer"~6 OR "staatssekretär* mahrer"~5 OR "harald mahrer"~4 OR "minister* rendi wagner"~5 OR "minister* wagner"~5 OR "bundesminister* rendi wagner"~5 OR "bundesminister* wagner"~5 OR "gesundheitsminister* wagner"~5 OR "gesundheitsminister* rendi wagner"~5 OR "frauenminister* wagner"~5 OR "frauenminister* rendi wagner"~5 OR "pamela rendi wagner"~5 OR "rendi wagner"~5 OR "minister* rupprechter"~5 OR "andrä rupprechter"~4 OR "landwirtschaftsminister* rupprechter"~5 OR "umweltminister* rupprechter"~5 OR "minister* schelling"~5 OR "finanzminister* schelling"~5 OR "hans jörg schelling"~4 "hans jörg schelling" OR "johann georg schelling" OR ("minister* sobotka"~5 NOT tschechi*) OR "innenminister* sobotka"~5 OR "wolfgang sobotka"~4 OR "van der bellen" OR bundespräsident OR strache* OR "matthias strolz"~4 OR "neos* strolz"~6 OR lunacek OR "peter pilz"~4 OR "ex grün* pilz"~6 OR "grüne* pilz"~6 OR "politiker* pilz"~5 OR "aufdecker* pilz"~5 OR "untersuchung* pilz"~5 OR "abgeordnete* pilz"~5 "parlament* pilz"~5 OR lunacek OR "rot grün" OR "schwarz grün" OR (("die grüne*" OR "der grüne*" OR "den grünen") NOT ("grün* rasen"~2 OR "grün* wiese*"~2)) OR ökopartei OR grünpolitiker* OR "grün* politiker*"~4 OR "tiroler grüne*" OR "grüne* tirol" OR "wiener grüne*" OR "grüne* wien" OR "grüne* burgenland" OR "burgenländ* grüne*" OR "grüne* kärnten" OR "kärntner* grüne*" OR "grüne* salzburg" OR "salzburger* grüne*" OR "grüne* oberösterreich" OR "oberösterreichische* grüne*" OR "grüne* steiermark" OR "steirische* grüne*" OR "grüne* vorarlberg" OR "vorarlberger grüne*" OR "grüne* niederösterreich*" OR "niederösterreichische* grüne*")
```

The data collection and the subsequent cleaning resulted in a final data set consisting of 46,205 articles. For further analysis, the data set was uploaded to the online content analysis platform AmCAT – see www.amcat.nl for more information and documentation.

4. Coding Procedure

4.1. Overview

We used Lucene syntax search strings to measure all variables. Wherever possible, we built on already developed search strings from the AUTNES 2013 Automatic Content Analysis. These were available for all parties included in our data set except the newly founded Pilz List, two main candidates (Heinz-Christian Strache, Matthias Strolz), some style variables

and all issues (for a complete overview of available strings see the documentation of AUTNES 2013 Automatic Content Analysis). New search strings were created for all new variables as well as to extend our issue search strings with issues arising after the 2013 election (e.g., Brexit). These search strings were then adapted, iterated on, and constantly tested against a set of articles. For sentiment analysis, we made use of two already existing dictionaries on sentiments in German language by Rauh and SentiWS.

For all variables except the issue and sentiment variables, the adaption procedure included adding and removing of search terms, adapting the word distances, and varying different sub-blocks and operators. After each adaption round, the search strings were tested for recall and precision against a set of 100 articles. For the most complex search strings (conflict and games framing) the R-package *quanteda* (Benoit et al., 2018[1]) was used to retrieve all words in a window of 5 words before and after the original search-terms. Those words were then listed in order of occurrence and the most frequent ones were added to the searchstring. This procedure was repeated until no significant change in performance (i.e., recall and precision) was achieved.

For the issue variables, more than 800 search strings were tested against an initial validation set. The results were then aggregated to the highest level (see Table 2 for an overview of the highest issue levels). A sample of false positives as well as false negatives on the aggregated level were checked both manually and automatically for similarities (sample n=100 for each issue). Following this, new search strings were created, and already existing search strings adapted.

Finally, we checked the sentiment dictionaries for possible confounders. This meant going manually through every word or word combination included in the dictionaries and controlling for possible overlaps with prominent candidate and party names in the 2017 national election. This procedure led to the removal of the word "*kurz*" from the dictionary

[1] Benoit, K., Watanabe, K., Wang, H., Nulty, P., Obeng, A., Müller, S., & Matsuo, A. (2018). quanteda: An R package for the quantitative analysis of textual data. *Journal of Open Source Software*, *3*(30), 774. https://doi.org/10.21105/joss.00774

by Rauh due to possible conflicts with the appearance of Sebastian Kurz, the main candidate of the ÖVP, in media reports.

The search strings as well as their recall and precision values are presented in the next subsections. A detailed account of the final validation procedure can be found in the following section (Section 5). These strings together with the meta data of the articles were used to code all variables on article level. *Table 2* provides an overview of all 53 variables included in the data set.

Table 2. Variables overview.

Variable	Label	Type
v0	Article ID	Metadata
v1	Headline	Metadata
v2	Byline	Metadata
v3_l1	Medium Level 1	Metadata
v3_l2	Medium Level 2	Metadata
v3_l3	Medium Level 3	Metadata
v4	Date	Metadata
v5	Page	Metadata
v6	Section	Metadata
v7	Word count	Metadata
v8	amcat uniqueID	Metadata
v9a	SPÖ	Party
v9b	ÖVP	Party
v9c	FPÖ	Party
v9d	Greens	Party
v9e	NEOS	Party
v9f	Pilz List	Party
v10a	Christian Kern	Candidate
v10b	Sebastian Kurz	Candidate
v10c	Heinz-Christian Strache	Candidate
v10d	Ulrike Lunacek	Candidate
v10e	Matthias Strolz	Candidate
v10f	Peter Pilz	Candidate
v11a	Economy, Budget and Taxes	Issue
v11b	Welfare State	Issue
v11c	Education and Culture	Issue
v11d	Security and Army	Issue
v11e	Foreign Policy	Issue
v11f	Europe	Issue
v11g	Infrastructure	Issue
v11h	Society	Issue
v11i	Environmental Policies	Issue
v11j	Immigration	Issue
v11k	Labour Market and Employment	Issue

Variable	Label	Type
v12	Conflict	Style
v13	Language of Games, Sports and War	Style
v14	Reference to EU	Other
v15	Social Media	Other
v16	Fake News	Other
v17	Polling Institutes	Other
v18	Silberstein-Affair	Other
v19a	Positive Sentiment - Rauh	Sentiment
v19b	Negative Sentiment - Rauh	Sentiment
v19c	Overall Sentiment - Rauh	Sentiment
v20a	Positive Sentiment - SentiWS	Sentiment
v20b	Negative Sentiment - SentiWS	Sentiment
v20c	Overall Sentiment - SentiWS	Sentiment
v21a	Personalization SPÖ	Personalization
v21b	Personalization ÖVP	Personalization
v21c	Personalization FPÖ	Personalization
v21d	Personalization Greens	Personalization
v21e	Personalization NEOS	Personalization
v21f	Personalization Pilz List	Personalization

4.2. Metadata

Metadata were derived from the source texts (digital newspaper articles, scraped news sites, or transcripts) automatically by AmCAT, where necessary manually corrected or recoded.

- ***Article ID* (v0):** ID number by which the article can be recognized and searched within AmCAT.
- ***Headline* (v1):** Headline of the news report, stored as a string variable.
- ***Byline* (v2):** The text below the headline in a news report, for example a sub-headline, location or an author, depending on the layout of the source text. Note that some articles do not come with a byline (n=13,867).
- ***Medium* (v3_l1-v3_l3):** Reported on three different levels. Level 1 (*v3_l1*) reports the type of media (TV, print, online), level 2 (*v3_l2*) aggregates media outlets to subclasses (e.g., Tabloids) and level 3 (*v3_l3*) reports the specific media outlet in which the article was published. For an overview on all included media outlets and their corresponding values and labels on each level, see *Table 3*.

Table 3. Medium (v3) - Level values and labels

L1 Value	L1 Label	L2 Value	L2 Label	L3 Value	L3 Label
1000	TV	1100	Public Service TV	1101	ZiB1
				1102	ZiB2
		1200	Commercial TV	1201	ATV
				1202	Pro7 – Puls 4
2000	Print	2100	Quality Print	2101	Der Standard
				2102	Die Presse & Die Presse am Sonntag
				2103	Salzburger Nachrichten
		2200	Tabloid Print	2201	Kronen Zeitung
				2202	Österreich
				2203	Heute
		2300	Mid-market Print	2301	Kurier
				2302	Kleine Zeitung
3000	Online	3100	Quality online	3101	derstandard.at
				3102	diepresse.com
				3103	salzburg.com/sn.at
		3200	Tabloid online	3201	krone.at
				3202	oe24.at
				3203	heute.at
		3300	Mid-market online	3301	orf.at
				3302	kurier.at
				3303	kleinezeitung.at

- ***Date* (v4):** The date on which the article was published.
- ***Page* (v5):** The page on which the article was published. Only reported for articles in print media (when available).
- ***Section* (v6):** Automatically coded variable on the section in which the article was published. Differentiating section names were harmonized to the following section labels in

- *Table **4***. A section was only assigned when possible. Sections are only available for print media. Overall, 21,835 articles do not have an assigned section.

Table 4. Section (v6) - Values and labels

Value	Label	Value	Label
1	Title Page	11	Science and Health
2	Domestic Politics	12	Sports
3	Topic of the Day	13	Letter to the Editor
4	Local News	14	Opinion Piece
5	Foreign Affairs	15	TV/Radio-Programs
6	Feuilleton	16	Events and Activities
7	Economy	17	Reportage and Features
8	Culture	33	Other
9	Communication and Media	99	Not assignable
10	Society		

- ***Word count* (v7):** Length of main text of articles, coded automatically.
- ***AmCAT unique ID* (v8):** Article IDs on online content analysis platform AmCAT (www.autnes.amcat.nl, Project: 3, Article Set ID: 5800).

4.3. Parties

The appearance in news reports of the main parties in the 2017 national election (SPÖ, ÖVP, FPÖ, Greens, NEOS, Pilz List) was coded with the following search strings. All party variables have the value 1 when the given party is mentioned in the news report, and 0 if it is not mentioned.

- ***SPÖ* (v9a):**

```
spo sp "partei sozialdemo*"~5 sozialdemokrat* "die roten"
"den roten" sozi sozialist*
```

[Precision: 0.91 | Recall: 0.97]

- ***ÖVP* (v9b):**

```
ovp vp volkspartei "liste kurz"~5 "neue* volkspartei" "die
schwarzen" "den schwarzen"
```

[Precision: 0.86 | Recall: 0.96]

- ***FPÖ* (v9c):**

```
fpo freiheitlich* fp "partei freiheitlich*"~5 "partei Strache*"~5
```

[Precision: 0.86 | Recall: 0.96]

- ***Greens* (v9d):**

```
("grune partei" "die grunen" "die grune" "der grunen" "der grune" "die grune partei" "grune alternative" okopartei grunpolitiker* "grun politiker*" "den grunen" "tiroler grune*" "wiener grune*" "grune* tirol" "grune* wien" "grune* burgenland" "burgenland* grune*" "grune* karnten" "karntner grune*" "grune* salzburg" "salzburger grune*" "grune* oberosterreich" "oberosterreichische* grune*" "grune* steiermark" "steirische* grune*" "grune* vorarlberg" "vorarlberger grune*" "grune* niederosterreich*" "niederosterreichische* grune*" "grune* spitzenkandidat*"~5 "grune* vorstandssprecher*"~5 "grune* landtagsabgeordnet*"~5 "grune* fpo*"~5 "grune* freiheitl*"~5 "grune* blau*"~5 "grune* spo"~5 "grune* sp"~5 "grune* sozialdemokrat*"~5 "grune* sozialist*"~5 "grune* rot*"~5 "grune* ovp"~5 "grune* vp"~5 "grune* volkspartei"~5 "grune* schwarz*"~5 "grune* neos"~5 "grune* kpo"~5 "grune* kommunist*"~5 "grune* chef*"~5 "grune* vorsitz*"~5 "grune* felipe"~5 "grune* lunacek"~5 "grune* sprecher*"~5 "grune* bundesvorstand*"~5 "grune* bundessprecher*"~5 "grune* landesvorstand*"~5 "grune* parlament*"~5 "grune* regierung*"~5 "grune* opposition*"~5 "grune* partei*"~5 "grune* bundnis*"~5 "grune* fraktion*"~5 "grune* klub*"~5 "grune* vize*"~5 "grune* burgermeister"~5 "grune* stadtrat*"~5 "grune* landesrat*"~5 "grune* bereichssprecher*"~5 "grune* referent*"~5 "grune* landespart*"~5 "grune* ortsgrupp*"~5) NOT("grune* (deutsch* cdu csu fdp jamaika spd afd ozdemir)"~7 "bundnis 90 grune*"~5 "grune* mark")
```

[Precision: 0.85 | Recall: 0.88]

- ***NEOS* (v9e):**

```
neos "das neue osterreich" "wahlplattform lif"~5
```

[Precision: 0.85 | Recall: 0.99]

- ***Pilz List* (v9f):**

```
"liste pilz"~10 "pilz zadic"~5 "pilz holzinger"~5 "pilz kolba"~5 "pilz zinggl"~5 "pilz roscher"~5 "pilz chelucci"~5 "pilz nasswetter"~5 "pilz cox"~5 "pilz werthner"~5 "pilz bißmann"~5 "pilz rossmann"~5 "pilz schroeder"~5 "liste peter pilz"~10
```

[Precision: 0.84 | Recall: 0.81]

4.4. Main candidates

The appearance in news reports of the main candidates in the 2017 national election was coded with the following search strings. A detailed account on the validation of these search strings can be found in the following section on the validation procedure. All variables have the value 1 when the given candidate is mentioned in the news report, and 0 if he/she is not mentioned.

- **Christian *Kern* - *SPÖ* (v10a):**

```
"kern christian"~4 "kern spö"~5 "kern kanzler*"~5 "kern sozialdemokrat*"~5 "kern spitzenkandidat*"~5 "bundeskanzler* kern"~5
```

[Precision: 0.99 | Recall: 0.85]

- **Sebastian *Kurz* - *ÖVP* (v10b):**

```
"kurz ovp"~5 "sebastian kurz"~4 "minister kurz"~5 "bundesminister* kurz"~5 "europa* kurz"~5 "integration* kurz"~5 "aussenminister* kurz"~5 "ausseres* kurz"~5 "kurz volkspartei"~5 "kurz vp"~5
```

[Precision: 0.98 | Recall: 0.87]

- ***Heinz*-Christian *Strache* - *FPÖ* (v10c):**

```
strache*
```

[Precision: 0.99 | Recall: 0.99]

- ***Ulrike Lunacek - Greens*** **(v10d):**

```
lunacek*
```

[Precision: 0.99 | Recall: 0.99]

- ***Matthias Strolz - NEOS*** **(v10e):**

```
"strolz matthias"~5 "strolz neo*"~10 "strolz obmann*"~10 "strolz spitzenmann*"~10 "strolz parteichef*"~10 "strolz boss*"~10 "strolz spitzenkandidat*"~10 "strolz vorsitz*"~10 "strolz chef*"~10 "strolz bundessprecher*"~10
```

[Precision: 0.98 | Recall: 0.90]

- ***Peter Pilz – Pilz List*** **(v10f):**

```
"pilz peter"~15
```

[Precision: 0.94 | Recall: 0.92]

4.5. Issues

We measure the appearance of issues by aggregating search string results on the finest level to the most general level that is reported in our data set. Issues were only validated on the most general level. Therefore, we recommend validating issue search strings on more fine-grained levels when needed. Due to the extensive number of search strings, only precision and recall values as well as the amount of search strings used for one issue are reported in this documentation. For all issues, the variable takes the value 1 if the issue is mentioned and 0 if it is not mentioned in the article.

- ***Economy, Budget and Taxes*** **(v11a):**

Aggregates the results of 113 search strings, covering the following topics: Economic policies, liberalization, deregulation, globalization, financial markets and banks, business policies, agriculture, tourism, state budget, government debt, and taxation policies.

[Precision: 0.61 | Recall: 0.44]

- ***Welfare State* (v11b):**

Aggregates the results of 101 search strings, covering the following topics: Welfare policies and spending, health system, care, pensions, family, and children support, housing, and sports. For details on the validation procedure of this specific issue, see the section on validation (Section 5).

[Welfare State (pt. 1), general - Precision: 0.83 | Recall: 0.85]

[Welfare State (pt. 2), specific - Precision: 0.73 | Recall: 0.59]

- ***Education and Culture* (v11c):**

Aggregates the results of 66 search strings, covering the following topics: Educational policies, schools, universities, research, vocational education, adult education, and cultural policies.

[Precision: 0.54 | Recall: 0.72]

- ***Security and Army* (v11d):**

Aggregates the results of 71 search strings, covering the following topics: Crime, penal system, police, terrorism, and military policies.

[Precision: 0.64 | Recall: 0.65]

- ***Foreign Policy* (v11e):**

Aggregates the results of 17 search strings, covering the following topics: International cooperation, international and bilateral relations and issues on war and peace.

[Precision: 0.76 | Recall: 0.73]

- ***Europe* (v11f):**

Aggregates the results of 78 search strings, covering the following topics: EU integration, EU treaties, EU extensions and exits, EU policies, EU participation and the Euro.

[Precision: 0.67 | Recall: 0.61]

- ***Infrastructure* (v11g):**

Aggregates the results of 38 search strings, covering the following topics: Road, rail and air traffic, shipping, energy policies, telecommunication, digitalization, and infrastructure investments and policies.

[Precision: 0.84 | Recall: 0.75]

- ***Society* (v11h):**

Aggregates the results of 76 search strings, covering the following topics: Gender equality, marriage, sexual orientation, abortion, bioethics, religion, censorship, handling of Austria's National Socialist Past, far-right extremism, anti-Semitism, patriotism, nationalism, minority rights and media policies.

[Precision: 0.80 | Recall: 0.68]

- ***Environmental Policies* (v11i):**

Aggregates the results of 25 search strings, covering the following topics: Environmental protection, climate change and animal welfare.

[Precision: 0.70 | Recall: 0.74]

- ***Immigration* (v11j):**

Aggregates the results of 61 search strings, covering the following topics: Immigration, refugees, asylum, integration of immigrants, Islam, racism and anti-racism, citizenship policies, and the handling of immigrants concerning the welfare system, labour market, education, and crime.

[Precision: 0.79 | Recall: 0.78]

- ***Labour Market and Employment* (v11k):**

Aggregates the results of 44 search strings, covering the following topics: Labour market, unemployment, employee rights and welfare and consumer protection.

[Precision: 0.62 | Recall: 0.65]

4.6. Style

The following two search strings measure the appearance of certain stylistic elements in news reports. Both variables take the value 1 if the stylistic element is present, and 0 if not. A detailed account on the validation of these search strings can be found in the following section on the validation procedure.

- ***Conflict* (v12):**

Our operationalization of a conflict frame includes cases where at least one of the following indicator questions applies: (1) *Does the item reflect disagreement between parties, individuals, groups or countries?* OR (2) *Does one party, individual or group insult/offend/accuse another?* Note that we do not code cases that refer to two or more sides of the problem under our operationalization of a conflict frame. The following search string was used to measure conflict:

```
(((kritik* NOT("kaum kritik")) kritisier* kampfansage (streit* NOT("außer streit" "streitmacht" "kein* streit")) gestritt* gegner* unredlich* unfahig* konflikt konflikte (angriff* NOT("in Angriff nehmen" "in Angriff genommen")) "nicht akzept*"~5 attack* heftige* (protest* NOT(protestwähler)) offensive angreifen angegriffen "griff* an"~10 "greift an"~10 vollholler unvereinbarkeit* Eklat tumult* strittig* intervention* intervenier* "wehr* gegen"~5 "vorwurf gegen*"~10 "warf vor"~10 "wirft vor"~10 frontalangriff* (widerspricht NOT("widerspricht nicht")) (widersprach NOT("widersprach nicht")) (widersprochen NOT("nicht widersprochen")) angegriffen* gezerr* schimpf* geschimpft*) AND NOT (ostukraine donbass syrien* syrisch* agypt* djihad* iran* sudan* sudsudan* nordkorea* un uno saudi* staatengemeinschaft* sicherheitsraat* terroris* kaida qaida pakistan sudan* jemen* libyen lybisch* somali* eritre* kongo* elfenbeinkust* ivor* ivoire hamas hisbollah miliz gotteskrieger taliban* is "islamische* staat" isis))
```

[Precision: 0.80 | Recall: 0.55]

- ***Language of Games, Sports and War* (v13):**

This variable captures whether a news report features terms of games, sports or war to describe actions and interaction between political actors or the election itself. The

following search string was used to measure the appearance of language of games, sports and war:

```
((agressiv* durchpeitsch* zerstor* "greift an"~5 "griff an"~5
"finish" vs "Aufholjagd" "Nachspiel" show* kill* endspurt
schlagabtausch verspiel* zweikampf* flugelkampfe kraftakt
"presch* vor" dreikampf* angriff* angegriff* angreif*
vergeltung* nahkampf* wortgefecht* scharmutzel* gefecht*
krieg kriege kriegerisch* arsenal* giftschrank* aufmarsch*
gegenschlag* uberraschungsangriff* uberraschungsschlag*
armada bombig* bombadier* bombardier* explosiv* propaganda*
neutralisier* konter kontern gekontert kontert*
konterattack* konterangriff* feldzug* sprengkraft
sprengstoff sprengt* schlachtfeld* offensiv* defensiv*
hinterhalt* schlacht hauptquartier (kampf* NOT("kampf*
tranen"~5)) gekampft* kanone* (schiess* NOT(schiesserei*))
geschoss* zuschlagen zugeschlagen* zuschlug zuschlagt
"anschlag* auf" kommando* geheimwaffe* belager* krise*
regierungskrise* koalitionskrise* attacke* wahlschlacht*
himmelfahrtskommando* kreuzzug* blitzkrieg* verteidigung
verteidigt* schutzengraben* geschutz mobilisier* erober*
vormarsch* bekampf* keule verbotskeule richtungskampf*
putsch* widerstand* feind* mitstreiter* rebellisch* aufstand
aufruhr* intervention* revolte* palastrevolte* gerangel
krawall* scharmutzel* konfrontation* konfrontier*
frontalangriff* kampagne* kampagnisier* aufbegehr*
einschiess* eingeschoss* entscharf* stichel* heckenschutze*
abschiess* abgeschoss* (abschuss NOT("abschuss tier*"~20))
querschiess* querschuss* querschutze* (front* NOT(frontex
"front national" frontalkollision "front* auto*"~10 "front*
fahrzeug*"~10 "front* unfall*"~10 "front* kolli*"~10))
koalitionsfront* regierungsfront* oppositionsfront*
grabenkampf* gelandegewinne vorstoss* deckung* storfeuer*
durchbruch* abruck* ruckzugsgefecht* verluste* ruckzug
vordringen* anfuhrer "fuhrung ausbauen"~5 "in fuhrung"~5
positionspapier* bollwerk* manover* taktik* taktisch*
wahltaktisch* strategie* strategisch* wahlstrategi*
kapitulation* kapitulier* "flucht nach vorn" fluchtet*
stormanover* giftpfeil* blockad* blockier* mobilisier*
rekrutier* flankier* ruckendeck* "rucken gedeckt"
mitstreiter* bollwerk* schlachtschiff* machtkampf*
kampfkandidat* kampfabstimm* uberflugel* vorbeizieh*
vorbeigezog* tauchstation* abtauch* abgetaucht* favorit
favoritin aussenseiter* outsider* geheimfavorit* rochade*
personalrochade* rochier* todesspirale* foul* finale*
angezahlt* angeschlagen* triumph* pyhrrus* steilpass*
steilvorlage* matchwinner* eigentor* powerplay arena
wahlkampfarena duell* kanzlerduell* gladiator* auferstehung*
```

```
opfergang* aufstachel* aufgestachel* fair* unfair* gewinn* sieger* verlierer* schlappe uberrunde* bruchlandung* angeschlagen* angezahlt* startschuss* startrampe* startlocher* "sturm auf" "ansturm auf" "war room" "aus volle* rohr*" "ins abseits" "im abseits" "rote karte" "auflage verwandel*" "vorlage verwandel*" "fest im sattel" "sicher im sattel" "auf der strecke bleib*" "blieb auf der strecke" "auf der strecke geblieb*" "kopf an kopf" "ins ziel" "ins spiel" "in stellung" "stellung ein*" "stellung beziehen" "bezog* stellung" "stellung halten" "stellung gehalten" "hielt stellung"~3) NOT (syrien* assad syrisch* agypt* djihad* iran* sudan* sudsudan* nordkorea* un uno staatengemeinschaft* sicherheitsraat* terroris* kaida qaida pakistan* sudan* jemen* libyen lybisch* somali* eritre* kongo* elfenbeinkust* ivor* ivoire hamas hisbollah miliz gotteskrieger taliban* terrormiliz "IS-terrormiliz" "IS" "kampfjets" bombardier "ISIS" terroranschlag* "terror anschlag" "todlich* anschlag" terrorakt terrorgefahr terrorregime* terrormiliz terrorwarnstufe* terrororganisation*))
```

[Precision: 0.62 | Recall: 0.63]

4.7. Other

Next to the variables capturing actors, issues and stylistic features, the appearance of the following five items was coded. All variables take the value 1 if the news report contains the specific item, and 0 if not.

- ***Reference to EU*** **(v14):**

Measures whether a news report contains an explicit reference to the European Union itself, EU policies or EU decision-making using the following search string:

```
(eu "(eu europaisch*) kommission*"~5 "(eu europaisch*) parlament*"~5 "(eu europaisch*) rat*"~5 kommissionsprasident* "kommission* prasident*"~4 ezb "europaisch* zentralbank*"~3 ratsprasident* eugh "(eu europaisch*) gerichtshof*"~3 "eu kommissar*"~5 "relocation programm" "(eu europaisch*) staatsanwalt*"~3 "eu (innenminister ausenminister) rat"~5 "eu (innenministerrat ausenministerrat)"~5 "(eu europaisch*) ausenbeauftragt*"~3 "ausschuss* der regionen" "(eu europaisch*) rechnungshof"~3 "(eu europaisch*) wirtschaft* sozialauschuss"~5 "(eu europaisch*) burgerbeauftragt*"~3 "(eu europaisch*) (ceta ttip)"~10 "(eu europaisch*) grenzschutzagentur*"~4 euratom
```

```
frontex "eu gipfel*" eurogruppe* "euro gruppe*" "eu (deal richtlinie* gesetz* vorschrift* vorgabe* regelung* genehmigung regel geregelt recht rechtlich* primarrecht* sekundarrecht* abkommen freihandelsabkommen grundverordnung behorde* agentur* landwirtschaftsfonds regionalentwicklungsfonds sozialfonds)"~6 "europaisch* (deal richtlinie* gesetz* vorschrift* vorgabe* regelung* genehmigung regel geregelt recht rechtlich* primarrecht* sekundarrecht* abkommen freihandelsabkommen grundverordnung behorde* agentur* landwirtschaftsfonds regionalentwicklungsfonds sozialfonds)"~8)
```

[Precision: 0.94 | Recall: 0.83]

- ***Social Media* (v15):**

Measures whether a news report contains an explicit reference to the Social Media, Social Media platforms or Social Media sources using the following search string:

```
(twitter* facebook* instagram* whatsapp* tweet* snapchat* pinterest* youtube* "social media*" "sozial* medien*" posting postet* gepostet getwittert) NOT("teilen share on facebook tweet" "whatsapp-service")
```

[Precision: 0.96 | Recall: 0.91]

- ***Fake News* (v16):**

Measures whether a news report contains an explicit reference to the role and extend of fake news or the usage of fake news by political actors using the following search string:

```
"fake news" falschmeldung "alternativ* fa:t*" postfakt* "post fa:t*" lügenpresse desinformation medienbashing "medien bashing"~3 "foto* manipul*"~5 "bild* manipul*"~5 "information* manipul*"~5 "gef:lscht information*"~5 "verf:lsch* information*"~5 "f:lschlich* information*"~5 "unkorrekt* information*"~5 lügenpresse* "lüge* presse"~5 "lüge* medien"~5 "lüge* medium"~5 "lüge* zeitung"~5 "lüge* orf"~5 "gelogen medien"~5 "gelogen medium"~5 "gelogen presse"~5 "gelogen zeitung"~5 "gelogen orf"~5 "unehrlich* medien"~5 "unehrlich* medium"~5 "unehrlich* presse"~5 "unehrlich* zeitung"~5 "unehrlich* orf"~5 "unwahr* medien"~5 "unwahr* medium"~5 "unwahr* presse"~5 "unwahr* zeitung"~5 "unwahr* orf"~5 "erfunden* medien"~5 "erfunden* medium"~5 "erfunden* presse"~5 "erfunden* zeitung"~5 "erfunden* orf"~5 "verlogen* medien"~5 "verlogen* medium"~5 "verlogen*
```

```
presse"~5 "verlogen* zeitung"~5 "verlogen* orf"~5
```

[Precision: 0.94 | Recall: 0.87]

- ***Polling Institutes* (v17):**

Measures whether a news report contains an explicit reference to a polling institute using the following search string:

```
umfrag* AND (ifes "integral meinungsforschung*" sora gallup imas "imas international" ogm "market institute" "unique research" "peter hajek" akonsult spectra "spectra markforschung" "research affairs" gfk)
```

[Precision: 1.00 | Recall: 0.94]

- ***Silberstein-Affair* (v18):**

Measures whether a news report contains an explicit reference to the Silberstein-Affair using the following search string:

```
silberstein OR „peter puller"~4
```

[Precision: 1.00 | Recall: 1.00]

4.8. Sentiment

To calculate sentiment-scores on article level, we used two existing dictionaries for German language: A sentiment dictionary for German political language by Christian Rauh (Rauh, 2018[2]) and the more general sentiment dictionary SentimentWortschatz/SentiWS (Remus, et al., 2010; v.2.0[3]). Both dictionaries attribute positive and negative sentiments to certain cue words. The overall score of negativity or positivity was normalized by the number of words of the given article. Note that numbers and punctation were removed from the texts as it is assumed that these do not bear any sentiment.

[2] Rauh, C. (2018). *Replication Data for: Validating a sentiment dictionary for German political language* (DRAFT VERSION). Harvard Dataverse. https://doi.org/10.7910/DVN/BKBXWD

[3] Remus, R., Quasthoff, U., & Heyer, G. (2010). *SentiWS-A Publicly Available German-language Resource for Sentiment Analysis.* LREC. http://www.lrec-conf.org/proceedings/lrec2010/pdf/490_Paper.pdf

For both dictionaries, we provide a separate normalized score of positivity and negativity together with an overall normalized sentiment score. The normalized positivity score theoretically ranges from 0 (no positive sentiment cues) to 1 (all words of the text contain positive sentiment), whereas the normalized sentiment score ranges from 0 (no negative sentiment cues) to -1 (all words of the text contain negative sentiment). As the overall normalized sentiment score is calculated as the sum of the two separate sentiment scores, it theoretically ranges from -1 to 1. As both dictionaries have already been validated by their creators, we did not conduct additional validation of sentiment scores. The following variables in the data set capture the sentiment of the news reports:

- ***Sentiment Rauh* (v19a-c):** Normalized sentiment scores calculated with the sentiment dictionary for German political language (Rauh, 2018). We provide information on sentiments in three ways: Positive sentiment – Rauh (*v19a*), Negative sentiment – Rauh (*v19b*) and Overall sentiment – Rauh (*v19c*).
- ***Sentiment SentiWS* (v20a-c):** Normalized sentiment scores calculated with the SentiWS dictionary (Remus, Quasthoff & Heyer, 2010; v.2.0). We provide information on sentiments in three ways: Positive sentiment – SentiWS (*v20a*), Negative sentiment – SentiWS (*v20b*) and Overall sentiment – Rauh (*v20c*).

4.9. Personalization

Personalization is operationalized as a continuous variable that represents the ratio of the summed mentions of a party's main candidate to the sum of main candidate and party mentions per article. It therefore ranges between 0 (only party is mentioned) to 1 (only main candidate is mentioned). Note that personalization is not coded (NA) if neither party nor main candidate is mentioned in a media report. As party and main candidate search strings were already validated separately (see Section 4.3 and 4.4), we do not provide validation for personalization scores. We provide personalization scores for the following six parties and their respective main candidates:

- ***Personalization SPÖ* (v21a)**
- ***Personalization ÖVP* (v21b)**

- ***Personalization FPÖ*** **(v21c)**
- ***Personalization Greens*** **(v21d)**
- ***Personalization NEOS*** **(v21e)**
- ***Personalization Pilz List*** **(v21f)**

5. Validation Procedure

5.1. Overview

Following face validity checks during the search string creation and adjustment process (see Section 4.1) we performed a final validation of our automatic methods by comparing their results to manual coding. This comparison results in a following binary contingency table:

	Manual = 1	**Manual = 0**	
Automatic = 1	True Positive (A)	False Positive (B)	*Predicted Positives* (A+B)
Automatic = 0	False Negative (C)	True Negative (D)	*Predicted Negatives* (C+D)
	Real Positives (A+C)	*Real Negatives* (B+D)	

Based on this comparison, we calculated to measures for validation: Precision and recall. Precision refers to the ratio of true positive results to all predicted positive results of a search string [P=A/(A+B)]. Recall refers to the ratio of true positive results of a search string to all real positives [R=A/(A+C] (Powers, 2011[4]).

For most variables we were able to draw on the manually coded data from the AUTNES Manual Content Analysis of the Media Coverage 2017 (n=9841). However, as some variables were not coded in the manual analysis or their operationalisation and coding differed between the manual and automatic content analysis, we conducted additional validation coding. We followed the proposed validation scheme of Stryker et al. (2006[5]). Based on the precision and recall values of our face validity checks we calculated the minimum number of coded articles needed to sufficiently validate our search strings. This

[4] Powers, D. M. (2011). Evaluation: From Precision, Recall and F-Measure to ROC, Informedness, Markedness & Correlation. *Journal of Machine Learning Technologies*, 2(1): 37–63. https://arxiv.org/abs/2010.16061

[5] Stryker, J. E., Wray, R. J., Hornik, R. C., & Yanovitzky, I. (2006). Validation of Database Search Terms for Content Analysis: The Case of Cancer News Coverage. Journalism & Mass Communication Quarterly, 83(2), 413–430. https://doi.org/10.1177/107769900608300212

resulted in a validation data set of 1000 articles (AmCAT set ID: 3770) to validate the missing items. The four student coders that coded our additional validation set were extensively trained, receiving in total three days of coder training. To test intercoder reliability, at the end of the training we conducted reliability tests that can be found in the following subsection (Section 5.2).

The issue variable v11b (Welfare State) is an exceptional case, as this variable is validated by combining the manual coding data from AUTNES 2017 and the additional validation data. As the AUTNES Manual Content Analysis only included specific welfare issues (housing, pensions, health and foster care, protection of minors, children and family), we had to recode the issues concerning welfare spending and general welfare policies. For v11b we therefore provide two precision and recall values. Note that for precision we used results of the corresponding sub-issue search strings, whereas results of all search strings on Welfare State was used to calculate recall for both validation sets.

Furthermore, note that we refrained from validating all the sentiment variables, as the two sentiment dictionaries have already been validated by their creators. Personalization variable did not require any validation, as the party and candidate variables used to calculate personalization scores, are validated on their own. Due to the low occurrence of the variables v16 (Fake News), v17 (Polling Institutes) and v18 (Silberstein-Affair), only the results of face validity checks on a reduced sample that was likelier to include these variables are reported. For these checks, we took a random sample of 50 articles from the search results of each string in the reduced samples to calculate precision, and a random sample of 100 articles of the total population of the reduced samples to calculate recall. Finally, note that we do not provide validation for metadata variables (v0-v8), as these variables were captured automatically from the information provided by the respective data sources.

Table 5 shows how each variable was validated. If it was validated via data from AUTNES Manual Content Analysis of the Media Coverage 2017, the corresponding variable(s) of the manual data set is given.

Table 5. Validation procedure

Variable	Label	Validation data	AUTNES manual variable
v9a	SPÖ	AUTNES manual	v14a (recoded)
v9b	ÖVP	AUTNES manual	v14c (recoded)
v9c	FPÖ	AUTNES manual	v14e (recoded)
v9d	Greens	AUTNES manual	v14g (recoded)
v9e	NEOS	AUTNES manual	v14i (recoded)
v9f	Pilz List	AUTNES manual	v14k (recoded)
v10a	Christian Kern	AUTNES manual	v14b (recoded)
v10b	Sebastian Kurz	AUTNES manual	v14d (recoded)
v10c	Heinz-Christian Strache	AUTNES manual	v14f (recoded)
v10d	Ulrike Lunacek	AUTNES manual	v14h (recoded)
v10e	Matthias Strolz	AUTNES manual	v14j (recoded)
v10f	Peter Pilz	AUTNES manual	v14l (recoded)
v11a	Economy, Budget and Taxes	AUTNES manual	v11h
v11b	Welfare State	AUTNES manual & Additional Validation	v11b, v11c, v11f, v11k
v11c	Education and Culture	AUTNES manual	v11l
v11d	Security and Army	AUTNES manual	v11g
v11e	Foreign Policy	Additional Validation	
v11f	Europe	AUTNES manual	v11e
v11g	Infrastructure	Additional Validation	
v11h	Society	Additional Validation	
v11i	Environmental Policies	AUTNES manual	v11d
v11j	Immigration	AUTNES manual	v11j
v11k	Labour Market and Employment	AUTNES manual	v11a
v12	Conflict	Additional Validation	
v13	Language of Games, Sports and War	Additional Validation	
v14	Reference to EU	Additional Validation	
v15	Social Media	Additional Validation	
v16	Fake News	Face Validity	
v17	Polling Institutes	Face Validity	
v18	Silberstein-Affair	Face Validity	

5.2. Intercoder Reliability

Table 6 shows the intercoder reliability of the additional validation set (n=1000), depicts both Krippendorff's Alpha and Percentage Agreement for each variable. For intercoder reliability on all other variables, consult the documentation of the AUTNES Manual Content Analysis of Media Coverage 2017 (DOI: 10.11587/IYGIBZ).

Table 6. Intercoder reliability - Additional validation

Variable	Label	Krippendorff's Alpha	Percentage Agreement
v11b (pt. 1)	Welfare State (general)	0.466	0.82
v11e	Foreign Policy	0.474	0.91
v11g	Infrastructure	0.698	0.88
v11h	Society	0.600	0.82
v12	Conflict	0.634	0.73
v13	Language of Games, Sports and War	0.604	0.68
v14	Reference to EU	0.903	0.96
v15	Social Media	0.700	0.96

5.3. Results

Table 7. Validation - Precision and Recall

Variable	Label	Precision	Recall
v9a	SPÖ	0,91	0,97
v9b	ÖVP	0,86	0,96
v9c	FPÖ	0,86	0,96
v9d	Greens	0,85	0,88
v9e	NEOS	0,85	0,99
v9f	Pilz List	0,84	0,81
v10a	Christian Kern	0,99	0,85
v10b	Sebastian Kurz	0,98	0,87
v10c	Heinz-Christian Strache	0,99	0,99
v10d	Ulrike Lunacek	0,99	0,99
v10e	Matthias Strolz	0,98	0,90
v10f	Peter Pilz	0,94	0,92
v11a	Economy, Budget and Taxes	0.61	0.44
v11b (pt. 1)	Welfare State (general)	0.83	0.78
V11b (pt. 2)	Welfare State (specific)	0.73	0.59
v11c	Education and Culture	0.54	0.72
v11d	Security and Army	0.64	0.65
v11e	Foreign Policy	0.76	0.73
v11f	Europe	0.67	0.61
v11g	Infrastructure	0.84	0.75
v11h	Society	0.80	0.68
v11i	Environmental Policies	0.70	0.74
v11j	Immigration	0.79	0.78
v11k	Labour Market and Employment	0.62	0.65
v12	Conflict	0.80	0.55
v13	Language of Games, Sports and War	0.62	0.63
v14	Reference to EU	0.94	0.83
v15	Social Media	0.96	0.91
v16	Fake News	0.94	0.87
v17	Polling Institutes	1.00	0.94

Variable	Label	Precision	Recall
v18	Silberstein-Affair	1.00	1.00

6. Data Set

In this section, we present a short overview over all variables in our dataset. To maintain clarity, recoded variables are shown in their most aggregated form. In addition to frequency tables, a short nominal description is given for each variable.

6.1. v6 Section

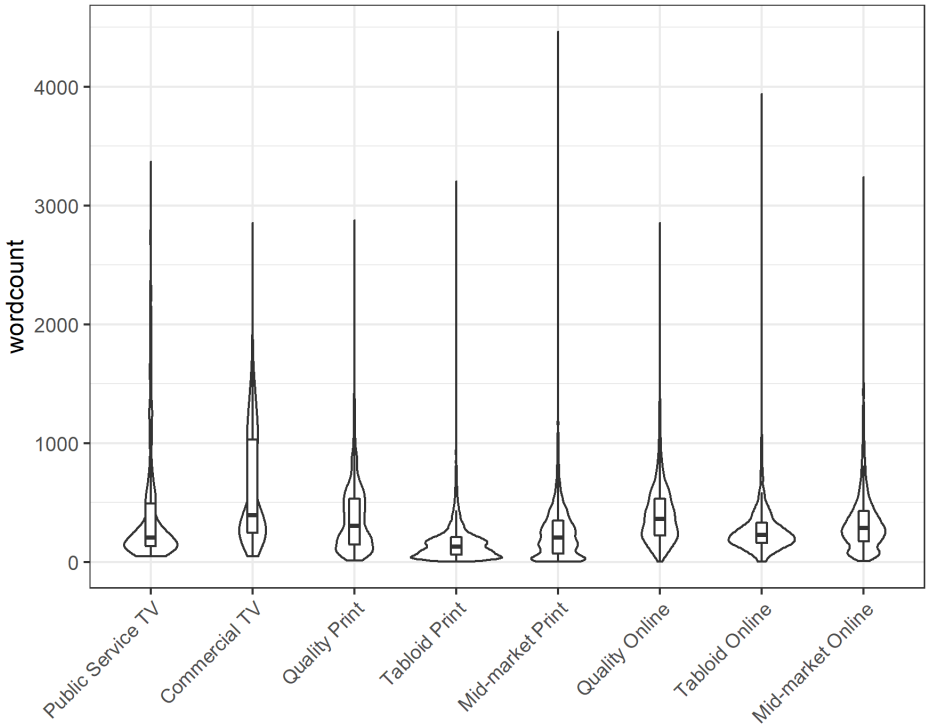
This variable was coded automatically and harmonized. It identifies the section in which an article was published.

	Public Service TV	Commercial TV	Quality Print	Tabloid Print	Mid-market Print	Quality Online	Tabloid Online	Mid-market online	Overall
Title Page			8.47	2.59	2.49				2.52
Domestic Politics			22.38	25.28	24.09				14.62
Topic of the day			8.33	3.17	2.21				2.59
Local news			1.75	39.18	23.73				14.67
Foreign Affairs			5.00	0.39	3.10				1.51
Feuilleton			7.71	6.81	16.95				6.19
Economy			6.79	1.26	3.67				2.13
Culture			1.29	0.85	2.52				0.90
Communication & Media			0.70		1.41				0.38
Society			0.05	0.87	2.13				0.63
Science & Health			0.16	0.21					0.08
Sports			0.77	1.47	0.87				0.66
Letter to the editor			0.39	3.25	2.10				1.28
Opinion piece			8.90		0.31				1.53
TV/Radio-Programs			2.05	1.26	0.57				0.76
Events and Activities				0.09	0.70				0.16
Reportage and feature			0.22	0.99	0.26				0.34
Other			2.57	4.56	1.16				1.79
Missing values	100	100	22.46	7.78	11.72	100	100	100	47.26
N	675	588	7614	11676	8720	5054	3359	8517	46203

Note: Percentage totals may be subject to rounding errors associated with individual components

6.2. v7 Word count

This variable was coded automatically. It counts the number of words each article contains.



	Mean	SD	Median
Public Service TV	419.97	519.09	208
Commercial TV	632.01	475.02	393
Quality Print	367.55	279.35	304
Tabloid Print	176.02	223.51	130
Mid-market Print	249.61	239.60	204
Quality online	408.70	275.43	361
Tabloid online	280.29	236.66	230
Mid-market online	345.00	281.34	287
Overall	295.02	280.12	224

6.3. v9a-f Parties

These variables measure whether a party is or is not mentioned in an article.

	Public Service TV	Commercial TV	Quality Print	Tabloid Print	Mid-market Print	Quality Online	Tabloid Online	Mid-market online	Overall
v9a – SPÖ	46.22	57.48	51.31	40.56	43.76	56.15	51.98	53.88	48.23
v9b – ÖVP	45.03	60.20	50.00	39.92	39.81	55.36	53.32	54.26	47.20
v9c – FPÖ	28.74	38.10	26.94	25.63	24.09	32.51	38.31	30.03	28.25
v9d – Greens	22.81	30.27	25.11	19.30	19.06	27.90	20.90	24.21	22.37
v9e – NEOS	16.88	27.72	14.07	9.74	9.56	18.30	13.81	14.96	12.95
v9f – Pilz List	7.55	14.97	5.42	2.64	2.95	9.36	4.35	5.37	4.75
N	675	588	7614	11676	8720	5054	3359	8517	46203

Note: Percentage totals may be subject to rounding errors associated with individual components

6.4. v10a-f Main candidates

These variables measure whether a main candidate is or is not mentioned in an article.

	Public Service TV	Commercial TV	Quality Print	Tabloid Print	Mid-market Print	Quality Online	Tabloid Online	Mid-market online	Overall
v10a – Kern	20.14	36.90	19.58	16.80	14.84	20.66	27.60	17.59	18.55
v10b – Kurz	24.44	44.56	25.83	19.15	18.62	28.39	33.73	23.40	23.41
v10c – Strache	11.70	28.40	9.81	12.06	8.58	12.09	20.39	10.38	11.53
v10d – Lunacek	6.81	20.92	4.96	4.59	3.83	7.14	7.92	6.94	5.70
v10e – Strolz	7.26	21.26	4.07	3.71	3.93	6.07	6.61	5.92	4.96
v10f – Pilz	10.22	18.20	8.64	5.83	5.65	7.14	7.14	6.20	6.79
N	675	588	7614	11676	8720	5054	3359	8517	46203

Note: Percentage totals may be subject to rounding errors associated with individual components

6.5. v11a-k Issues

These variables measure whether a specific issue is mentioned in a media report. Note that an article can include more than one issue.

	Public Service TV	Commercial TV	Quality Print	Tabloid Print	Mid-market Print	Quality Online	Tabloid Online	Mid-market online	Overall
v11a – Economy, Budget and Taxes	24.00	34.01	32.34	17.74	22.86	33.58	22.15	30.79	25.87
v11b – Welfare State	15.26	26.53	20.78	11.55	14.22	22.30	15.54	19.89	16.82
v11c – Education and Culture	19.11	18.88	23.35	12.27	18.26	23.47	13.78	20.56	18.28
v11d – Security and Army	20.59	23.30	18.87	11.85	13.92	19.81	19.74	18.72	16.38
v11e – Foreign Policy	6.81	8.50	8.73	3.01	4.81	8.63	5.42	7.24	5.99
v11f – Europe	8.44	13.10	8.41	3.02	4.46	9.00	6.82	8.55	6.34
v11g – Infrastructure	4.59	6.80	10.82	8.77	11.02	11.06	8.04	12.00	10.24
v11h – Society	12.30	18.88	15.88	6.17	9.20	18.34	13.37	13.11	11.73
v11i – Environmental Policies	5.33	9.35	7.76	5.81	6.54	7.74	6.22	9.04	7.14
v11j – Immigration	19.55	33.33	22.47	11.86	13.94	23.92	22.83	20.55	18.11
v11k – Labour Market and Employment	12.59	22.62	17.27	8.89	11.78	18.18	10.99	15.70	13.47
N	675	588	7614	11676	8720	5054	3359	8517	46203

Note: Percentage totals may be subject to rounding errors associated with individual components

6.6. v12-v18 Style and other variables

These variables measure whether an article features a certain element of style (conflict, language of games, sports and war) or if other specific topics (Reference to EU, social media, fake news, polling institutes, Silberstein-affair) are mentioned in the article.

	Public Service TV	**Commercial TV**	**Quality Print**	**Tabloid Print**	**Mid-market Print**	**Quality Online**	**Tabloid Online**	**Mid-market online**	**Overall**
v12 – Conflict	28.00	35.71	35.05	21.43	26.00	38.58	34.30	34.59	30.05
v13 – Language of Games, Sports and War	37.19	54.59	44.04	35.85	35.80	51.11	47.66	43.37	41.36
v14 – Reference to EU	14.07	16.16	16.80	6.91	9.78	17.10	12.21	15.44	12.38
v15 – Social Media	8.30	10.88	8.80	5.67	6.19	11.71	19.08	10.85	8.98
v16 – Fake News	1.19	0.17	1.00	0.31	0.63	1.09	0.80	0.68	0.68
v17 – Polling Institutes	1.63	2.21	0.70	1.36	0.85	0.69	2.50	1.00	1.11
v18 – Silberstein-Affair	5.04	7.65	4.79	3.87	3.10	8.43	13.01	7.10	5.70
N	675	588	7614	11676	8720	5054	3359	8517	46203

Note: Percentage totals may be subject to rounding errors associated with individual components

6.7. v19a-c Sentiment (Rauh)

These variables measure positive, negative and overall sentiment on article level based on the sentiment dictionary for German political language by Christian Rauh (Rauh, 2018[6]).

		Mean	SD	Min	Max
v19a – Positive Sentiment	Public Service TV	0.04943	0.02295	0	0.18519
	Commercial TV	0.05024	0.02556	0	0.22500
	Quality Print	0.04948	0.02600	0	0.25000
	Tabloid Print	0.04946	0.02604	0	0.30769
	Mid-market Print	0.04921	0.02537	0	0.28125
	Quality online	0.04891	0.02480	0	0.33333
	Tabloid online	0.05027	0.02590	0	0.26667
	Mid-market online	0.04940	0.02475	0	0.26667
	Overall	0.04941	0.02548	0	0.33333
v19b – Negative Sentiment	Public Service TV	-0.03883	0.02221	-0.16667	0
	Commercial TV	-0.03841	0.02423	-0.16667	0
	Quality Print	-0.03728	0.02416	-0.25000	0
	Tabloid Print	-0.03723	0.02436	-0.39286	0
	Mid-market Print	-0.03744	0.02357	-0.29630	0
	Quality online	-0.03701	0.02315	-0.23684	0
	Tabloid online	-0.03611	0.02317	-0.24242	0
	Mid-market online	-0.03705	0.02396	-0.31579	0
	Overall	-0.03718	0.02386	-0.39286	0
v19c – Overall Sentiment	Public Service TV	0.01059	0.03449	-0.16667	0.18519
	Commercial TV	0.01184	0.03894	-0.16667	0.22500
	Quality Print	0.01220	0.03723	-0.25000	0.25000
	Tabloid Print	0.01223	0.03775	-0.39286	0.30769
	Mid-market Print	0.01177	0.03634	-0.29630	0.28125
	Quality online	0.01189	0.03569	-0.23684	0.33333
	Tabloid online	0.01416	0.03644	-0.21053	0.26667
	Mid-market online	0.01234	0.03617	-0.26316	0.26667
	Overall	0.01223	0.03676	-0.39286	0.33333

[6] Rauh, C. (2018). *Replication Data for: Validating a sentiment dictionary for German political language* (DRAFT VERSION). Harvard Dataverse. https://doi.org/10.7910/DVN/BKBXWD

6.8. v20a-c Sentiment (SentiWS)

These variables measure positive, negative, and overall sentiment on article level based on the sentiment dictionary SentimentWortschatz/SentiWS (Remus et al., 2010; v.2.0[7]).

		Mean	SD	Min	Max
v20a – Positive Sentiment	Public Service TV	0.00419	0.00419	0.00006	0.04069
	Commercial TV	0.00428	0.00396	0.00005	0.02398
	Quality Print	0.00416	0.00430	0.00001	0.06881
	Tabloid Print	0.00410	0.00418	0.00002	0.06007
	Mid-market Print	0.00414	0.00438	0.00002	0.08300
	Quality online	0.00407	0.00429	0.00003	0.09056
	Tabloid online	0.00415	0.00427	0.00003	0.06223
	Mid-market online	0.00418	0.00477	0.00003	0.12690
	Overall	0.00414	0.00436	0.00001	0.12690
v20b – Negative Sentiment	Public Service TV	-0.01161	0.00975	-0.11716	-0.00155
	Commercial TV	-0.01211	0.01128	-0.11298	-0.00137
	Quality Print	-0.01244	0.01189	-0.20431	-0.00081
	Tabloid Print	-0.01248	0.01150	-0.18881	-0.00100
	Mid-market Print	-0.01229	0.01127	-0.17655	-0.00106
	Quality online	-0.01198	0.01099	-0.19509	-0.00071
	Tabloid online	-0.01195	0.01072	-0.13059	-0.00061
	Mid-market online	-0.01218	0.01146	-0.17655	-0.00090
	Overall	-0.01227	0.01138	-0.20431	-0.00061
v20c – Overall Sentiment	Public Service TV	-0.00742	0.01066	-0.11616	0.02800
	Commercial TV	-0.00783	0.01241	-0.11218	0.01346
	Quality Print	-0.00827	0.01284	-0.20358	0.02609
	Tabloid Print	-0.00838	0.01252	-0.18824	0.03346
	Mid-market Print	-0.00815	0.01236	-0.17455	0.04198
	Quality online	-0.00792	0.01186	-0.19359	0.03094
	Tabloid online	-0.00780	0.01159	-0.12996	0.02082
	Mid-market online	-0.00799	0.01234	-0.17455	0.09159
	Overall	-0.00813	0.01235	-0.20358	0.09159

[7] Remus, R., Quasthoff, U., & Heyer, G. (2010). *SentiWS-A Publicly Available German-language Resource for Sentiment Analysis.* LREC. http://www.lrec-conf.org/proceedings/lrec2010/pdf/490_Paper.pdf

6.9. v21a-f Personalization

These variables measure how personalized media coverage of a certain party is. It is computed by taking the ratio of main candidate mentions to the sum of mentions of each main candidate and his/her party.

v21a Personalization SPÖ

	Mean	SD	Min	Max
Public Service TV	0.163	0.292	0	1
Commercial TV	0.165	0.296	0	1
Quality Print	0.176	0.297	0	1
Tabloid Print	0.168	0.288	0	1
Mid-market Print	0.167	0.284	0	1
Quality online	0.167	0.286	0	1
Tabloid online	0.173	0.296	0	1
Mid-market online	0.163	0.281	0	1
Overall	0.169	0.288	0	1

v21b Personalization ÖVP

	Mean	SD	Min	Max
Public Service TV	0.230	0.294	0	1
Commercial TV	0.190	0.283	0	1
Quality Print	0.221	0.308	0	1
Tabloid Print	0.222	0.299	0	1
Mid-market Print	0.224	0.301	0	1
Quality online	0.221	0.300	0	1
Tabloid online	0.222	0.307	0	1
Mid-market online	0.227	0.304	0	1
Overall	0.223	0.302	0	1

v21c Personalization FPÖ

	Mean	SD	Min	Max
Public Service TV	0.196	0.305	0	1
Commercial TV	0.177	0.312	0	1
Quality Print	0.218	0.329	0	1
Tabloid Print	0.221	0.329	0	1
Mid-market Print	0.226	0.331	0	1
Quality online	0.223	0.326	0	1
Tabloid online	0.222	0.331	0	1
Mid-market online	0.207	0.318	0	1
Overall	0.218	0.327	0	1

v21d Personalization Greens

	Mean	SD	Min	Max
Public Service TV	0.025	0.105	0	1
Commercial TV	0.019	0.088	0	1
Quality Print	0.022	0.098	0	1
Tabloid Print	0.021	0.094	0	1
Mid-market Print	0.021	0.095	0	1
Quality online	0.020	0.093	0	1
Tabloid online	0.018	0.087	0	1
Mid-market online	0.020	0.092	0	1
Overall	0.021	0.094	0	1

v21e Personalization NEOS

	Mean	SD	Min	Max
Public Service TV	0.205	0.286	0	1
Commercial TV	0.167	0.253	0	1
Quality Print	0.168	0.258	0	1
Tabloid Print	0.168	0.259	0	1
Mid-market Print	0.167	0.263	0	1
Quality online	0.163	0.257	0	1
Tabloid online	0.169	0.257	0	1
Mid-market online	0.172	0.263	0	1
Overall	0.169	0.260	0	1

v21f Personalization Pilz List

	Mean	SD	Min	Max
Public Service TV	0.523	0.335	0	1
Commercial TV	0.424	0.351	0	1
Quality Print	0.551	0.362	0	1
Tabloid Print	0.528	0.363	0	1
Mid-market Print	0.517	0.361	0	1
Quality online	0.516	0.377	0	1
Tabloid online	0.532	0.371	0	1
Mid-market online	0.522	0.365	0	1
Overall	0.526	0.365	0	1